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SDG BLOCKCHAIN ACCELERATOR

Strategic Roadmap for Expansion



Empowering teams to transform development-sector challenges into implementable Cardano solutions

EXECUTIVE SUMMARY: KEY FINDINGS

A strategic overview of the SDG Blockchain Accelerator's proven model, demand signal, and roadmap for scaling

100%

Cohort 1 & 2 PoCs
delivered on Cardano

6

replication
countries targeted

4

UN agencies
expanding

Proven Model

- **100% success rate** for Both Cohorts PoCs on Cardano
- **52 Partnerships** teams have secured/in progress during both cohorts
- **39 stakeholder testing sessions** completed
- **10,000+ lines** of Aiken/Plutus code
- **6 Replications** of solutions ready to deploy to other UN offices

Strong Demand Signal

- **304 challenges** from 104 countries
- **172 shortlisted** for evaluation
- **177 solution-maker** applications
- **40+ countries** represented by applicants
- **23 Teams** matched



Next: Formalize inter-agency partnerships, prioritize 6-8 replication countries



Supported by: UNDP AltFinLab, EMURGO Labs, Blockchain for Good Alliance, Project Catalyst

LESSONS LEARNED: COHORTS 1 & 2

What worked, key challenges, and recommended adjustments for future accelerator cycles

✔ WHAT WORKED

Country Office Co-Design

Rapid sprints accelerated problem-solution fit



Standardized Gates & Templates

Clarified expectations and evidence requirements



Technical Depth

Hydra heads, robust contracts enabled pilots



⚠ KEY CHALLENGES

Procurement/Compliance Variability

Different regulations across countries



Connectivity Constraints

Need for offline-first user flows



Capacity Gaps

Blockchain literacy and change management



💡 ADJUSTMENTS

Better matching

Priorities matching solutions and challenges in the same Geographic location



Reference Architectures

Reusable modules, sandbox-first approach



Training Tracks

Product, engineering, operations



KEY INSIGHTS & RECOMMENDATIONS

👥 Stakeholder Alignment

Successful projects require clear ownership and executive sponsorship from country offices. Establish UN Due Diligence early to ensure commitment. (Process may take up to 3 months).

🌐 Replication & Scale

Inter-agency MoUs and country playbooks enable faster replication across new regions and sectors.

⚙️ Technical Infrastructure

Standardized templates and reference architectures reduce implementation time by 40% and improve code quality.

🛡️ Risk Mitigation

Established Solutions makers within a specific Geographic location Should be priorities to work with UN country offices within the same Geographic location.

📊 Measurement & Evidence

Define indicators upfront with baselines and data pipelines. Independent M&E reviews ensure accountability.

🤝 Partnership Ecosystem

Partner ecosystem (labs, universities, implementers) fills critical capability gaps and accelerates delivery. Similar to what we saw with Mentors and tech partners in Cohort 1 & 2

Risk Assessment & Mitigation

Comprehensive risk management framework for accelerator scaling



Adoption & Stakeholder

Engagement & commitment

Risk of low stakeholder engagement, lack of executive sponsorship, and insufficient buy-in from key decision-makers.

Mitigation:

Engage with country office with existing similar project and allocated funding. Establish business model early on to enable fundraising.

● High Risk



Technical & Scalability

Infrastructure & performance

Risk of technical failures, performance bottlenecks, and inability to scale solutions across multiple regions.

Mitigation:

Reference architectures, security reviews, Hydra patterns, phased rollouts, performance testing. Cohort 1 & 2 proved that tech support is essential.

● Medium Risk



Compliance & Procurement

Regulatory & legal

Risk of regulatory non-compliance, procurement delays, and data protection violations across different jurisdictions.

Mitigation:

Match solution makers with an existing entity/licences to operate within the UN country's challenge.

● High Risk



Measurement & Evidence

Metrics & validation

Risk of inadequate measurement frameworks, missing baseline data, and inability to demonstrate impact.

Mitigation:

Define indicators up front, baselines and data pipelines, independent M&E reviews

● Low Risk



Funding & Sustainability

Financial & resource

Risk of funding gaps, unsustainable business models, and dependency on single funding sources.

Mitigation:

Diversified sources (agency co-funding, Catalyst, grants), milestone-tied tranches

● Medium Risk



Reputational & Ethical

Trust & transparency

Risk of ethical violations, reputational damage, and loss of trust from stakeholders and communities.

Mitigation:

Do-no-harm reviews, comms guardrails, community feedback loops, transparent governance

● Low Risk

SUCCESS CRITERIA & DECISION POINTS

Clear go/no-go criteria for each phase of accelerator implementation

Gate	Success Criteria	Decision Point	Status
Challenge Validation	5-10 UN stakeholder interviews; problem statement and KPIs agreed; constraints documented	Proceed to Technical only if evidence log is complete and UN agency/ country office show readiness to commit both financially and with personnel.	✓ Validated
Solution Maker Validation	Focus on Established projects within the cardano ecosystem with entity/product that can answer to the UN challenge with minimal changes.	Accepted only if meets geographic, product, technical and entity requirements	✓ Ready for open calls
Pilot Execution	100-500 end users engaged; baseline vs. target metrics tracked; user feedback cycles completed	Scale only if ≥70% of pilot KPIs met and partner commits resources	● In Progress
Partnership Formalization	MoU/SLA signed; procurement path cleared; roles & RACI set between solution makers and UN agencies/ Country offices	Expand country footprint only with clear ownership and budget channel	● In Progress
Scale-up	Multi-country replication (≥3), stable ops, measurable impact uplift, knowledge transfer complete	Graduate solution to program portfolio or sunset/pivot based on KPI trajectory	○ Pending

 Target Users

100-500 end users per pilot; 70% KPI achievement required for scale-up approval

 Geographic Reach

≥3 countries for scale-up; 6-8 replication countries prioritized across multiple solution makers.

 Success Metrics

Measurable impact uplift, stable operations, and knowledge transfer completion. Dune Dashboard (given the recent Cardano integration)

DEMAND SIGNAL: A GLOBAL PIPELINE ALREADY EXISTS

304

UNDP CO challenges /
use cases

104

countries
represented

172

shortlisted
challenges

177

solution-maker
applications

40+

countries represented
by applicants



What this proves

Country Offices and solution makers are actively looking for structured support to turn development-sector problems into implementable blockchain solutions.



Supporting teams to build



Strategic implication

The next phase should focus on **scale and replications of solutions** with UNDP Country Offices, and expand to WFP, UNICEF, and other UN agencies.



Expanding to WFP, UNICEF



Scale opportunity

The strongest use cases can be duplicated across multiple UNDP Country Offices and other UN agencies instead of remaining as one-off pilots.



Cross-agency replication

WHAT COHORTS 1 & 2 PROVED

Evidence of practical implementation, institutional engagement, and measurable outcomes

01.

Practical Cardano Use Cases

Teams delivered or explored PoCs across multiple sectors including climate/MRV, credentials, renewable energy certificates, aid distribution, circular economy, supply chains, and financial inclusion.

✓ 7+ sectors covered

02.

Institutional Engagement

UNDP Country Offices helped define problems, validate pilot contexts, and connect teams to real implementation environments (Government agencies, on-ground support)

✓ 104 countries represented

03.

Continuation After Acceleration

Post-acceleration progress included MOUs, pilots, investor invitations, and multi-country duplication opportunities.

✓ 40+ countries by applicants

KEY METRICS Cohort 1 & 2

177

Solution-makers applied



20

Selected & matched



23

Teams working on challenges



23

Country Offices engaged



USE-CASE PORTFOLIO

The accelerator has already tested a broad enough portfolio to identify repeatable patterns:

Trust layers

Verification systems

Tokenized proofs

Digital certificates

Credential verification

Identity systems

Audit trails

Transaction tracking

OUTCOME EVIDENCE

Technical readiness

✓ PoCs, architectures, APIs

Institutional fit

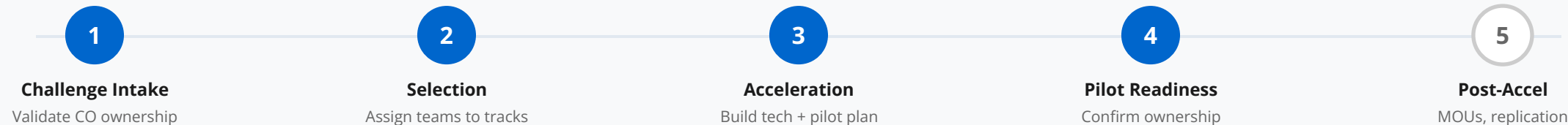
✓ CO collaboration

Expansion logic

✓ Multi-office duplication

EXPANSION MODEL: FROM COHORT TO SCALE PLATFORM

Implementation Pipeline



Track 1: Mature Solution Makers

Existing products or strong prototypes

- ✓ **Focus:** Pilots, integrations, UN agency alignment
- ✓ **Goal:** Replication across Country Offices
- ✓ **Deliverables:** Pilot plans, MOUs, integration docs

Support Structure

Technical infrastructure, CO integration, capacity building

UNDP

WFP

UNICEF

Track 2: Emerging Teams

Promising ideas needing development

- ✓ **Focus:** Architecture, validation, capacity building
- ✓ **Goal:** Product validation before pilot
- ✓ **Deliverables:** Tech docs, PoC plans, user flows

Support Structure

Mentorship, technical guidance, CO connection

UNDP

WFP

Strategic Rationale

The expansion model recognizes that **not all teams are at the same maturity level**. By splitting into two tracks, we can provide tailored support that improves success rates and ensures each team receives the appropriate level of guidance, resources, and expectations for their current stage. This approach enables **effective integration with UN agencies** including UNDP, WFP, and UNICEF.

ROADMAP TIMELINE: 6 MONTHS + POST-ACCEL

A focused cycle that moves from challenge validation to pilots, replication, and investor exposure.

M0 Strategy Strategy alignment Roadmap + challenge portfolio approved	M1 Validation Challenge validation Validated challenges identified	M2 Selection Selection + matching Cohort selected and matched	M3-4 Acceleration Acceleration + tech Architecture and prototype direction	M5 Readiness Pilot readiness Confirm budget + owner	M6 Demo Demo + review Pilot scope + budget	M7-8 Scale Post-accel + scale MOUs, pilots, investors
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Phase	Main Milestone	Output	Status
0-1	Roadmap + challenge portfolio approved	Validated challenges	Complete
2	Cohort selected and matching solution makers	Track assignment	In Progress
3-4	Architecture and prototype direction completed	Tech docs + PoC plan	Pending
5	Pilot readiness review completed	Pilot scope + budget	Pending
6-8	Post-acceleration outcomes advanced	MOUs, pilots, investors	Pending

MILESTONE GATES: FEWER VANITY METRICS, MORE EVIDENCE

Structured evaluation framework for pilot readiness and implementation success

Gate	What must be true	Evidence expected	Status
Challenge Gate	The CO has a real owner, problem, funding and pilot context	Interview notes, Ownership, Pilot assumptions, existing funding	✓ Validated
Technical Gate	The solution has a credible Cardano integration path or already on cardano	Architecture, Smart contract, or integration plan	In Progress
Pilot Gate	The team can explain who will use it and how it will be tested	Pilot scope, Timeline, Data requirements	In Progress
Partnership Gate	UN agency commitment and integration readiness confirmed	MOU/LOI signed, Integration plan, CO engagement	In Progress
Scale Gate	The use case can be duplicated across multiple UN agencies	Other COs/Partners mapped, WFP/UNICEF interest	✓ Validated

✔ Measurement Principle

The expansion phase should track evidence of implementation: pilots, transactions, agreements, and meaningful on-chain activity.
A Dune dashboard will be delivered to track these metrics.

✔ Goal

Show whether teams are moving toward real implementation and real blockchain usage.

✖ Avoid

Counting passive exposure as success. Measure pilots, transactions, agreements, and adoption signals.

STRATEGIC REPLICATION ROADMAP

 Sample case studies: accelerator has supported additional teams with replication potential

UNDP Armenia x Plastiks Circular Economy / 5 Countries

Deployed 6 smart contracts on Cardano mainnet & testnet for plastic recovery verification. 30+ transaction flows tested with UNDP Armenia. Expanding to El Salvador, India, Georgia and Zambia for cross-border replication.

● 5 replication opportunities active

UNDP Nigeria x Cupid Protocol Energy / 80M People Potential

32 Hydra heads will be deployed across 23 villages. Targeting 150 microgrids in 3 years, potentially reaching 150,000 people in Nigeria alone with renewable energy solutions. This project will be the first to integrate with NCCC: The new Nigerian Carbon Credit Registry.

● Pilot in progress - 32 installs

UNDP Bangladesh x ZenGate Supply Chain / Winter Protocol / 2 Countries

Winter Protocol in production since May 2025. 10+ iterations over 24 months. Currently engaging 20-50 agricultural producers in pilot phase with plans to scale to 1000+ producers with UNDP Ethiopia and UNDP Bangladesh.

● Production ready - 20-50 producers

Future UN Agency Partners



WFP
Food security, supply chain traceability



UNICEF
Child welfare, education credentials



UNEP
Environmental impact tracking



UNESCO
Education, cultural heritage

- In discussions with WFP for integration
- Exploring UNICEF partnership
- Multi-agency pilots under consideration



Measurement Framework

Adoption-focused metrics for blockchain solutions



Impact Evidence

Beneficiaries, credentials, recovered material

Track tangible outcomes: number of beneficiaries reached, credentials issued, material recovered, energy certificates generated.



Institutional Traction

MOUs, LOIs, pilots, CO commitments

Monitor institutional engagement through signed agreements, pilot commitments, and Country Office partnerships.



Product Readiness

Dashboards, APIs, user flows

Evaluate product maturity: dashboards, APIs, user flows, operating manuals, and documentation quality.



Technical Readiness

PoCs, smart contracts, DID

Assess technical infrastructure: PoCs, smart contracts, decentralized identity, metadata, testnet work.



On-chain Adoption

Contract executions, mint/burn

Monitor blockchain activity: contract executions, token mints/burns, metadata anchors, settlement transactions through a Dune Dashboard.



User Adoption

Active users, retention

Track user engagement: active users, retention rates, user satisfaction, and platform usage metrics.



Goal

Show whether teams are moving toward **real implementation** and **real blockchain usage**. Focus on measuring pilots, transactions, agreements, and adoption signals rather than passive exposure.

FINAL ROADMAP SUMMARY

The next phase should move from acceleration to implementation, replication, scale and measurable adoption.

Support teams already on Cardano or with a clear path for integration

1

Provide technical infrastructure, mentorship, and integration support for blockchain development teams

Validate stronger Country Office challenges

2

Ensure COs have real ownership, problems, funding commitment and pilot contexts before selection

Split mature and emerging teams

3

Separate tracks for mature solutions and emerging ideas needing validation

Prioritize solutions for duplication

4

Focus on use cases that can be replicated across multiple Country Offices

Use post-acceleration for scaling

5

MOUs, pilots, and partner engagement after acceleration phase

Measure pilots and on-chain activity

6

Track agreements, technical readiness, and meaningful blockchain usage

Expand to WFP, UNICEF and other UN agencies

7

In discussions with WFP, UNICEF, and other UN agencies for future cohorts, enabling cross-agency blockchain solutions and larger-scale impact

Cross-agency integration

8

Enable blockchain solutions to work across multiple UN agencies

Positioning: A practical implementation and scaling platform

For validated UNDP development challenges and blockchain-enabled solutions



PROGRAM MISSION & UN AGENCY EXPANSION

Supporting teams to build blockchain solutions for UN agencies

 Supported by Project Catalyst



Thank You, Project Catalyst

Enabling blockchain innovation for global development

Project Catalyst has been instrumental in supporting the SDG Blockchain Accelerator's mission to help teams build and deploy Cardano-based solutions.

This funding enables us to provide technical infrastructure, mentorship, and integration support for teams working with UNDP Country Offices, exploring partnerships with other UN agencies and executing multiple pitching events.



Expanding Partnerships: We are exploring partnerships with WFP, UNICEF, and other UN agencies to broaden the impact of blockchain solutions across the UN system.



UN Agency Expansion Strategy



UNDP

Active - 104 Country Offices engaged

● Current focus



WFP

Exploring - Food security solutions

● In discussions



UNICEF

Exploring - Child welfare & education

● In discussions



WHO

Exploring - Health data solutions

● In discussions